

EARN'M DropBox Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Dacian

August 15, 2024

Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for EARN'M on EARN'M DropBox. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated August 15, 2024, was delivered to EARN'M, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from EARN'M or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 13 days, the Cyfrin team conducted an audit on the EARN'M DropBox code provided by EARN'M. The findings consist of 1 Critical, 2 Medium & 2 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	EARN'M DropBox
Repository	dropbox-smart-contracts
Commit	f1d6baf1646b...
Audit Timeline	July 29th - Aug 14th 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	0	0	0
Medium Risk	2	2	0
Low Risk	2	1	1
Informational	7	7	0
Gas Optimizations	25	25	0
Total	37	36	1

6 Audit Scope

The following contracts together with associated interfaces and dependencies were included in the scope for this audit:

```
src/DropBox.sol
src/DropBoxFractalProtocol.sol
```

A new contract added during the audit was also audited:

```
src/VRFHandler.sol
```

The audit was conducted using a "collaborative agile" methodology with the client fixing issues soon after they were found and the auditor daily updating to the latest code version.

7 Risk Classification

	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

EARN'M's `DropBox` protocol is a new product similar to the previous `MysteryBox` product which was also audited by Cyfrin.

`DropBox` is a rewards program where users receive reward codes which are used to claim `ERC721` drop boxes of various tiers. Each tier allows users to redeem their drop boxes for a specific amount of `EARNM ERC20` tokens associated with that tier; the longer a user waits to redeem the more tokens they get for their drop box.

Box tiers are assigned using Chainlink's Verifiable Random Function (VRF) API v2.5.

Outside the protocol users are generally free to trade `ERC721` drop boxes amongst each other or on NFT marketplaces, however the `NFT` is burned in the same transaction at which the `EARNM` token redemption occurs.

The protocol also integrates with Limit Break's Creator Token Standards (CTS) library for royalty payments on NFT marketplaces.

9 Attestation

Cyfrin attests that it performed a security review of the EARN'M DropBox code provided by EARN'M.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at `https://github.com/Cyfrin/cyfrin-audit-reports`. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-08-15-cyfrin-earnm-dropbox-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to EARN'M has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=ba85ce5c689ef1484ff1110042dafc58fe6d3296be6a254e24918b5aa93c4314
F=2024-08-15-cyfrin-earnm-dropbox-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.